Enterprise SIEM Lab: Wazuh Cluster with Custom Alert Dashboards

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GitHub: https://github.com/jatinjangid2104/wazuh-siem-lab

Project Description

This project deploys a Wazuh SIEM cluster to aggregate logs from multiple endpoints, detect suspicious activity with custom rules mapped to MITRE ATT&CK, and visualize alerts on dashboards. A Jupyter notebook documents the deployment, defines the custom rules, checks cluster health through the Wazuh API, and analyzes alerts with Python.

Dataset

This project does not use a downloadable dataset. Its data is the live alert stream produced by the Wazuh cluster from the lab's monitored hosts and simulated attacks.

- **Live mode:** the notebook queries the Wazuh indexer (`wazuh-alerts-*`).
- **Sample mode (default):** if no cluster is reachable, the notebook generates synthetic alerts so it runs anywhere.

Dataset link: N/A (lab-generated logs). Wazuh documentation: https://documentation.wazuh.com

Technologies Used

- Wazuh (manager, indexer, dashboard, agents)
- Sysmon (Windows telemetry)
- Python 3: `requests`, `pandas`, `numpy`, `matplotlib`
- Jupyter Notebook
- MITRE ATT&CK framework

Repository Contents

File	Description
`JatinJangid_WazuhSIEMLab.ipynb`	Main project notebook (code)
`requirements.txt`	Python dependencies
`JatinJangid_ProjectReport.docx`	Full project report
`README.md`	This file

The notebook also writes `local_rules.xml` (custom rules) and `alert_dashboard.png` (charts) when run.

Setup and Run

1. Clone or download this repository.
2. Create a virtual environment and install dependencies:

```
python -m venv venv
source venv/bin/activate      # Windows: venv\Scripts\activate
pip install -r requirements.txt
```

3. Launch the notebook:

```
jupyter notebook JatinJangid_WazuhSIEMLab.ipynb
```

4. Run all cells. By default it uses sample data and needs no Wazuh cluster.

Using a live Wazuh cluster (optional)

In the notebook, set `LIVE = True` and update `API_URL`, `API_USER`, `API_PASS`, `INDEXER_URL`, `IDX_USER` and `IDX_PASS`. Do not commit real credentials.

Key Information

- **Custom rules:**
  - `100100`: SSH brute force (T1110)
  - `100110`: new local user created (T1136.001)
  - `100120`: encoded PowerShell command (T1059.001)
- **Deployment:** Wazuh was installed with the official assisted installer (indexer, server master and worker, dashboard).
- **Tested behaviors:** SSH brute force, user creation, encoded PowerShell and file integrity changes, simulated from a Kali VM.

- **Future work:** active response, Suricata network logs, vulnerability detection, threat intel enrichment.

Disclaimer

This is an educational lab. Run attack simulations only on systems you own or have permission to test.